

Paper-1

Lucknow Cantonment Board
(X-Ray Technician)

1. A micron is _____ times smaller than a millimetre.
(a) 250 (b) 500
(c) 1000 (d) 2000
2. G-CSF is produced by *A*
(a) Lymphocytes
(b) Stromal cells of the marrow
(c) Granulocytes
(d) Stem cells
3. Acid phosphatase is demonstrated by an azo dye coupling techniques that depends upon the hydrolysis of a substrate containing: *Chem*
(a) a-Napthol phosphate
(b) NAD
(c) NADPH
(d) G6PD
4. Zenker's solution is recommended for the fixation of small pieces of which type of tissue. *Phy*
(a) Liver (b) Spleen
(c) Both a & b (d) None Option
5. Name the gas used in preparation of bleaching powder: *Chem*
(a) Chlorine (b) Hydrogen
(c) Nitrogen (d) None option
6. Heat will accelerate the process of:
(a) Mounting (b) Dehydration
(c) Demineralization (d) Clearing
7. Which endocrine gland secrete insulin and glucagon hormone?
(a) Adrenal gland (b) Thymus gland
(c) Pituitary gland (d) Pancreas
8. Which disease is caused due to deficient iodine diet?
(a) Goiter (b) Acromegaly
(c) Cretinism (d) Beri-beri
9. Cushing's syndrome is caused due to hyperactivity of
(a) Adrenal medulla (b) Adrenal cortex
(c) Thyroid gland (d) Parathyroid gland
10. Gilt-edged market means
(a) Bullion market
(b) Market of government securities
(c) Market of guns
(d) Market of pure metals
11. Which one of the following lipoprotein has maximum density?
(a) LDL (b) Chylomicrons
(c) HDL (d) RNA
12. _____ component of Nucleus has phosphoric acid and four nitrogenous base.
(a) Chromatin (b) RNA
(c) DNA (d) Nucleoli
13. Areolar tissue, fibrous tissue, yellow elastic, reticular, adipose, lymphoid tissue, cartilage, bone & teeth are example of which type of tissue.
(a) Epithelial tissue
(b) Muscular tissue
(c) Connective tissue
(d) Nervous tissue

14. Which of the following types of energy can be used to decompose water into its elements?
 - (a) Heat energy
 - (b) Light energy
 - (c) Chemical energy
 - (d) Electrical energy
15. Blood pressure & cardiac output is increased during
 - (a) Sleeping
 - (b) Eating
 - (c) Exercise
 - (d) Both a & c
16. _____ is the longest & largest bone of the skeleton.
 - (a) Femur
 - (b) Tibia
 - (c) Fibula
 - (d) None of these
17. Muscles having double nerve supply:
 - (a) Digastrics muscle
 - (b) Trapezius
 - (c) Adductor magnus
 - (d) All of the above
18. A dark purple compound used as an antiseptic and disinfectant is:
 - (a) Potassium nitrate
 - (b) Sodium thiosulphate
 - (c) Potassium permanganate
 - (d) Calcium phosphate
19. Which wave in ECG represents ventricular septal activity and is shown by inconspicuous downward deflection?
 - (a) Q wave
 - (b) R wave
 - (c) S wave
 - (d) T wave
20. On which factor, the blood pressure is dependent?
 - (a) Systolic force
 - (b) Cardiac output
 - (c) Peripheral resistance
 - (d) All of these
21. Which type of cell is increased in body fluid in bacterial infection?
 - (a) Neutrophil
 - (b) Basophil
 - (c) Eosinophil
 - (d) Macrophages
22. Which is not found in DNA?
 - (a) Guanine
 - (b) Uracil
 - (c) Adenine
 - (d) Thymine
23. In strip method, the color of the strip is changed to _____ when the urine contains albumin.
 - (a) Yellow
 - (b) Orange
 - (c) Red
 - (d) Green
24. Hormones that regulate salt and water balance, except:
 - (a) Calcitonin
 - (b) Vasopressin
 - (c) Parathyroid hormone
 - (d) 1, 25-dihydroxycholecalciferol
25. Which method is used for the estimation of hormones?
 - (a) Radioimmunoassay (RIA)
 - (b) Fluorimetric assay
 - (c) ELISA
 - (d) All of the above
26. Amino acids are building blocks for
 - (a) Proteins
 - (b) Carbohydrates
 - (c) Lipids
 - (d) DNA
27. In diabetes _____ level is raised.
 - (a) Blood urea
 - (b) Blood sugar
 - (c) Blood cholesterol
 - (d) Blood lipid
28. Oils and fats are obtained from –
 - (a) Animal sources
 - (b) Vegetable sources
 - (c) Synthesis
 - (d) Both animal and vegetable sources
29. Serum amylase is increased in
 - (a) Renal failure
 - (b) Obesity
 - (c) Diabetes
 - (d) Acute pancreatitis
30. Which of the following is most important in causing coronary artery disease?
 - (a) LDL
 - (b) HDL
 - (c) VLDL
 - (d) Triglycerides
31. Which protein is synthesized by platelet?
 - (a) Albumin
 - (b) Globulin
 - (c) Fibrinogen
 - (d) Collagen
32. Which blood cell is essential for blood clotting?
 - (a) Red blood cell
 - (b) White blood cell
 - (c) Platelets
 - (d) All of the above

33. The difference between isotopes of an element is due to the presence of a different number of:
(a) Protons (b) Neutrons
(c) Electrons (d) Photons
34. Permanent hardness of water is due to the presence of –
(a) Calcium bicarbonate
(b) Magnesium bicarbonate
(c) Calcium Sulphate
(d) Sodium bicarbonate
35. _____ is known as the powerhouse of the cell.
(a) Ribosome (b) Golgi apparatus
(c) Lysosomes (d) Mitochondria
36. Malaria infection can be transmitted by
(a) Anopheles mosquito bite
(b) Blood transfusion
(c) Transmission through placenta
(d) All of the above
37. OoK is equivalent to:
(a) 273°C (b) -273°C
(c) 0°C (d) 100°C
38. In Myelography the intra-theccal contrast injection is injected into which space:
(a) Extra-dural
(b) Intra-medullary
(c) Sub-dural
(d) Sub-arachnoid
39. Basic protein is:
(a) Albumin (b) Globulin
(c) Histone (d) Glutelin
40. Ferritin is stored in
(a) Liver (b) Spleen
(c) Gut (d) All of the above
41. _____ components are used as an amplifying device?
(a) Diode (b) Electric cell
(c) Transformer (d) Transistor
42. The following two amino acids are semi-essential.
(a) Arginine, histidine
(b) Cysteine, tyrosine
(c) Proline, phenylalanine
(d) Serine, threonine
43. The temperature at which a substance catches fire is called its:
(a) Melting point
(b) Boiling point
(c) Heat temperature
(d) Ignition temperature
44. Intraoperative radiation therapy (IORT) is used to treat an exposed tumor during cancer surgery. Which rays are used in this treatment?
(a) UV rays
(b) Infrared rays
(c) X-rays, gamma rays, and electron beams
(d) Neutrons
45. All of the following infectious diseases are transmitted by milk except
(a) Tuberculosis (b) ~~Hepatitis A virus~~
(c) Brucellosis (d) Salmonellosis
46. Cytology helps in the diagnosis of
(a) Malignancy
(b) Evaluation of endocrine disorder
(c) In assessment of hormonal activity
(d) All of the above
47. Ultrasonic sounds are the sound which has frequencies more than:
(a) 30 KHz (b) 50 KHz
(c) 20 KHz (d) 15 KHz
48. Syphilis is caused by
(a) Treponema pallidum
(b) Neisseria gonorrhoeae
(c) Human papilloma virus
(d) Human immunodeficiency virus
49. In zinc sulphate floatation technique, what percentage of zinc sulphate solution is used?
(a) 25% (b) 31%
(c) 33% (d) 35%
50. Carbon dating technique is used to determine _____.
(a) Age of fossils
(b) Age of plastic
(c) Weight of non-metals
(d) Volume of liquids

Answer and Explanation

1. **Ans. (c) 1000**
Exp. A micron (micrometer) is one thousandth of a millimeter.
2. **Ans. (b) Stromal cells of the marrow**
Exp. Granulocyte colony-stimulating factor (G-CSF) is primarily produced by the stromal cells of the bone marrow.
3. **Ans. (a) a-Napthol phosphate**
Exp. Acid phosphatase activity is demonstrated using azo dye coupling techniques.
4. **Ans. (b) Spleen**
5. **Ans. (a) Chlorine**
Exp. Chlorine gas is used in the preparation of bleaching powder.
6. **Ans. (b) Dehydration**
Exp. Heat accelerates the process of dehydration.
7. **Ans. (d) Pancreas**
Exp. The pancreas secretes insulin and glucagon hormones.
8. **Ans. (a) Goiter**
Exp. Goiter is caused due to a deficient iodine diet.
9. **Ans. (b) Adrenal cortex**
Exp. Cushing's syndrome is caused by the hyperactivity of the adrenal cortex.
10. **Ans. (b) Market of government securities**
Exp. Gilt-edged market refers to the market of government securities.
11. **Ans. (c) HDL**
Exp. HDL (High-Density Lipoprotein) has the maximum density among lipoproteins.
12. **Ans. (c) DNA**
Exp. DNA is composed of phosphoric acid and four nitrogenous bases.
13. **Ans. (c) Connective tissue**
Exp. Areolar tissue, fibrous tissue, and others listed are examples of connective tissue.
14. **Ans. (d) Electrical energy**
Exp. Electrical energy can be used to decompose water into its elements.
15. **Ans. (c) Exercise**
Exp. Blood pressure and cardiac output increase during exercise.
16. **Ans. (a) Femur**
Exp. The femur is the longest and largest bone in the human skeleton.
17. **Ans. (d) All of the above**
Exp. Digastrics muscle, trapezius, and adductor magnus muscles have double nerve supply.
18. **Ans. (c) Potassium permanganate**
Exp. Potassium permanganate is a dark purple compound used as an antiseptic and disinfectant.
19. **Ans. (a) Q wave**
Exp. The Q wave in an ECG represents ventricular septal activity and is shown by an inconspicuous downward deflection.
20. **Ans. (d) All of these**
Exp. Blood pressure is dependent on systolic force, cardiac output, and peripheral resistance.
21. **Ans. (a) Neutrophil**
Exp. Neutrophils increase in body fluid in bacterial infections.
22. **Ans. (b) Uracil**
Exp. Uracil is found in RNA, not in DNA.
23. **Ans. (c) Red**
Exp. In the strip method, the color of the strip changes red when the urine contains albumin.
24. **Ans. (a) Calcitonin**
Exp. Calcitonin does not regulate salt and water balance; it is involved in calcium homeostasis.
25. **Ans. (d) All of the above**
Exp. Radioimmunoassay (RIA), Fluorimetric assay, and ELISA are methods used for hormone estimation.
26. **Ans. (a) Proteins**
Exp. Amino acids are the building blocks of proteins.
27. **Ans. (b) Blood sugar**
Exp. In diabetes, the blood sugar level is raised.
28. **Ans. (d) Both animal and vegetable sources**
Exp. Oils and fats are obtained from both animal and vegetable sources.
29. **Ans. (d) Acute pancreatitis**
Exp. Serum amylase is increased in acute pancreatitis.
30. **Ans. (a) LDL**
Exp. LDL (Low-Density Lipoprotein) is most important in causing coronary artery disease.

31. Ans. (c) Fibrinogen

Exp. Platelets synthesize fibrinogen.

32. Ans. (c) Platelets

Exp. Platelets are essential for blood clotting.

33. Ans. (b) Neutrons

Exp. The difference between isotopes of an element is due to the presence of a different number of neutrons.

34. Ans. (c) Calcium Sulphate

Exp. Permanent hardness of water is due to the presence of calcium sulfate.

35. Ans. (d) Mitochondria

Exp. Mitochondria are known as the powerhouse of the cell.

36. Ans. (d) All of the above

Exp. Malaria infection can be transmitted by Anopheles mosquito bite, blood transfusion, and transmission through the placenta.

37. Ans. (b) -273°C

Exp. 0K (zero Kelvin) is equivalent to -273°C .

38. Ans. (d) Sub-arachnoid

Exp. In Myelography, the intra-thecl contrast injection is injected into the sub-arachnoid space.

39. Ans. (c) Histone

Exp. Histone is a basic protein.

40. Ans. (d) All of the above

Exp. Ferritin is stored in the liver, spleen, and gut.

41. Ans. (d) Transistor

Exp. Transistors are used as amplifying devices.

42. Ans. (a) Arginine, histidine

Exp. Arginine and histidine are semi-essential amino acids.

43. Ans. (d) Ignition temperature

Exp. The temperature at which a substance catches fire is called its ignition temperature.

44. Ans. (c) X-rays, gamma rays, and electron beams

Exp. X-rays, gamma rays, and electron beams are used in intraoperative radiation therapy (IORT).

45. Ans. (b) Hepatitis A virus

Exp. Hepatitis A virus is not transmitted by milk.

46. Ans. (d) All of the above

Exp. Cytology helps in the diagnosis of malignancy, evaluation of endocrine disorders, and assessment of hormonal activity.

47. Ans. (c) 20 KHz

Exp. Ultrasonic sounds have frequencies more than 20 KHz.

48. Ans. (a) *Treponema pallidum*

Exp. Syphilis is caused by *Treponema pallidum*.

49. Ans. (c) 33%

Exp. In the zinc sulfate floatation technique, a 33% zinc sulfate solution is used.

50. Ans. (a) Age of fossils

Exp. Carbon dating technique is used to determine the age of fossils.

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